PAPER_932: Blazar Ergosphere Phonon Resonance

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** blazar_jet_phonon.py (BlazarErgosphereResonance) **Calculator:** BlazarErgospherePhononResonanceCalc (CP4 #516) **CVW:** v2.0.0 compliant

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Abstract

We derive the ergosphere phonon resonance conditions for BL Lac and Mrk 421 type blazars within the UQFF framework. The key result is that SCm phonon modes, when Doppler-boosted by relativistic bulk Lorentz factors Gamma_bulk ~ 15, satisfy the superradiant condition omega_obs < Omega_H for spinning black holes with a >= 0.95. The extracted ergosphere energy couples to the 26-layer gravity framework via E_ergo = (a/2) M c^2 S_26^2 delta_D.

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1. Core Equations

Doppler factor:

$$\delta_D = \frac{1}{\Gamma_{\text{bulk}}(1 - \beta \cos \theta_{\text{obs}})}$$

where $\beta = \sqrt{1 - 1/\Gamma_{\text{bulk}}^2}$.

Horizon angular velocity:

$$\Omega_H = \frac{a \cdot c}{2 r_H}$$

where $r_H = \frac{r_S}{2}(1 + \sqrt{1 - a^2})$ and $r_S = 2GM/c^2$.

Superradiant condition:

$$\omega_{\text{obs}} = \omega_{\text{SCm}} \cdot \delta_D < \Omega_H$$

Ergosphere phonon energy:

$$E_{\text{ergo}} = \frac{a}{2} M c^2 \cdot S_{26}^2 \cdot \delta_D$$

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2. UQFF Integration

The `BlazarErgospherePhononResonanceCalc` (CP4 #516) computes delta_D, omega_obs, Omega_H, the superradiant flag, and E_ergo for arbitrary blazar parameters. The simulate() method sweeps over Gamma_bulk = [5, 10, 15, 20, 30] to map the transition into and out of the superradiant regime.

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3. Physical Significance

Blazar jets are among the most powerful persistent energy sources in the universe. The Blandford-Znajek mechanism extracts rotational energy from the ergosphere via magnetic field threading. The UQFF contribution is a phonon-mediated channel: SCm lattice modes in the vacuum near the horizon are Doppler-shifted into the superradiant band, enabling additional energy extraction that modulates the observed jet power.

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4. Source Data

- **File:** blazar_jet_phonon.py
- **Session:** 212
- **CP4 Class:** BlazarErgospherePhononResonanceCalc (#516)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Blandford, R.D. & Znajek, R.L. – Electromagnetic extraction of energy from Kerr black holes (1977)
3. Urry, C.M. & Padovani, P. – Unified Schemes for Radio-Loud AGN (1995)